VSL objective: what the symbols mean and why the formula holds

2 October 2026 — explanation of the accepted analytical contract

1. The quantity being minimized

Yes: the VSL slide's J is **mean complete time per unit traffic mass**. If $\mathcal{T}$ is total system time for the affected cohort, then

$$J = \frac{\mathcal{T}}{M} = \tau + \frac{M + (\alpha - 1)e(\tau)}{2\mu}, \qquad \tau = \frac{L}{v}, \qquad e(\tau) = pM \exp(-\lambda\tau).$$

It includes travel through the preparation zone and all subsequent time until exit from the modeled choke. It is neither the time of the last departure nor merely a queue-delay score relative to a free-flow baseline.

Symbol	Meaning	Units
M	Total continuous vehicle-equivalent mass in the cohort.	veh-eq
p	Initial fraction of that mass in the lane that ends.	dimensionless
L, v	Preparation-zone length and common advice speed. The chosen branch assumes travel at this speed.	km; km/h
τ	Common preparation-zone traversal time.	h
λ	Fixed fractional transfer rate from ending to continuing lanes.	h^{-1}
$e(\tau)$	Mass still in the ending lane when the cohort reaches the choke; it still needs to merge.	veh-eq
$c(\tau)$	Mass already in continuing lanes: $M - e(\tau)$.	veh-eq
α	Receiving work per unit residual merging mass, relative to one unit for continuing-lane mass. $\alpha > 1$.	dimensionless
W	Total receiving workload: $c + \alpha e$.	work
μ	Constant available receiving-work rate, not a fixed vehicle exit rate.	work/h
T	Time to discharge the batch after arrival: W/μ.	h
$\mathcal{T}, J$	Total cohort system time; that total divided by M.	veh-eq·h; h

The work unit is normalized so ordinary continuing-lane mass uses one unit per vehicle-equivalent. α is the *total* merge-work multiplier; $\alpha - 1$ is its extra part. Neither creates extra vehicles. $e(\tau)$ names a mass; exp denotes the exponential function. All units may instead use seconds, provided rates and speeds are converted consistently. The derivative holds $M, p, L, \lambda, \alpha, \mu$ fixed.

2. Why travel time plus half the clearance time?

First, the action changes what remains to merge. The admitted fixed transfer branch satisfies

$$\frac{\mathrm{d}e}{\mathrm{d}t} = -\lambda e, \qquad \frac{\mathrm{d}c}{\mathrm{d}t} = +\lambda e, \qquad e(0) = pM.$$

Paired transfer conserves $e + c = M$. Longer traversal gives more opportunity for early transfer, leaving $e(\tau) = pM \exp(-\lambda\tau)$ at the choke. Preparation-zone receiving is assumed nonbinding on this branch.

Second, remaining merging mass costs additional receiving work. Continuing-lane mass costs one work unit; remaining merging mass costs α. Hence

$$W = c + \alpha e = (M - e) + \alpha e = M + (\alpha - 1)e, \qquad T = \frac{W}{\mu}.$$

The continuing lanes do not bypass the scarce receiving resource. Both classes consume it, under homogeneous proportional batch service. Their exit rates are

$$q_c = \frac{\mu c}{W}, \qquad q_e = \frac{\mu e}{W}, \qquad q = q_c + q_e = \frac{\mu M}{W} = \frac{M}{T}, \qquad q_c + \alpha q_e = \mu.$$

The last identity is work accounting; actual exiting mass is q, not μ. The homogeneous rule preserves composition as the batch drains.

Third, integrate the traffic still in the system. In this analytical reduction all mass starts together at time zero, travels for τ, then becomes a batch at the choke. If $N(t)$ is remaining cohort mass and u is time since batch arrival,

$$N(t) = M \quad (0 \le t < \tau), \qquad N(\tau + u) = M\left(1 - \frac{u}{T}\right) \quad (0 \le u \le T).$$

The area is a rectangle during travel followed by a triangle during discharge:

$$\mathcal{T} = \int_0^{\tau+T} N(t)\,\mathrm{d}t = M\tau + \int_0^T M\left(1 - \frac{u}{T}\right)\mathrm{d}u = M\tau + \frac{MT}{2}.$$

Dividing by M gives

$$\boxed{J = \tau + \frac{T}{2} = \underbrace{\tau}_{\text{travel}} + \underbrace{\frac{M}{2\mu}}_{\text{ordinary batch discharge}} + \underbrace{\frac{(\alpha-1)e(\tau)}{2\mu}}_{\text{additional merge work}}}.$$

The factor $1/2$ is a consequence of linear discharge, *not* a universal queueing identity. The last mass exits at $\tau + T$; the mean is $\tau + T/2$. Common action-independent prefix or downstream time can be added as a constant.

Illustration, not calibration. With $M = 100$ veh-eq, $e = 20$ veh-eq, $\alpha = 2$, $\mu = 10$ work/s and $\tau = 30$ s, $W = 120$ work and $T = 12$ s. Thus $J = 36$ s and $\mathcal{T} = 3600$ veh-eq·s; the last departure is at 42 s.

3. Deriving and interpreting the speed derivative

Suppressing the fixed hidden-state index, the complete dependence on the action is

$$v \longrightarrow \tau(v) = L/v \longrightarrow e(\tau) = pM\exp(-\lambda\tau) \longrightarrow W \longrightarrow J.$$

Step 1: differentiate remaining merging mass with respect to time.

$$\frac{\mathrm{d}e}{\mathrm{d}\tau} = -\lambda e(\tau).$$

Step 2: differentiate the mean-time objective with respect to time. Since M, α, μ are fixed,

$$\frac{\mathrm{d}J}{\mathrm{d}\tau} = 1 + \frac{\alpha-1}{2\mu}\frac{\mathrm{d}e}{\mathrm{d}\tau} = 1 - \frac{(\alpha-1)\lambda e(\tau)}{2\mu}.$$

Step 3: convert traversal-time sensitivity to speed sensitivity.

$$\frac{\mathrm{d}\tau}{\mathrm{d}v} = -\frac{L}{v^2}, \qquad \boxed{\frac{\mathrm{d}J}{\mathrm{d}v} = \frac{\mathrm{d}J}{\mathrm{d}\tau}\frac{\mathrm{d}\tau}{\mathrm{d}v} = -\frac{L}{v^2}\left[1 - \frac{(\alpha-1)\lambda e(\tau)}{2\mu}\right].}$$

Equivalently, direct differentiation gives $\mathrm{d}e/\,\mathrm{d}v = \lambda e L/v^2 > 0$: faster advice leaves more mass to merge at the choke. The cost derivative is the negative travel-time derivative plus the positive receiving-work derivative.

What the bracket means. An additional second of preparation time costs one second of mean travel time. To first order, it transfers λe additional mass per unit time and saves $(\alpha-1)\lambda e/(2\mu)$ units of mean choke time per unit preparation time.

- If this saving is less than one, travel cost dominates: $\mathrm{d}J/\,\mathrm{d}\tau > 0,\ \mathrm{d}J/\,\mathrm{d}v < 0$. Increasing speed locally helps.
- If the saving exceeds one, merge-work relief dominates: $\mathrm{d}J/\,\mathrm{d}\tau < 0,\ \mathrm{d}J/\,\mathrm{d}v > 0$. Reducing speed locally helps.
- Equality is the interior stationary condition, if reached within the allowed speed interval. Since $\mathrm{d}^2J/\,\mathrm{d}\tau^2 = (\alpha-1)\lambda^2 e/(2\mu) > 0$, it is the unique minimum in traversal time. Otherwise the relevant boundary action is optimal.

Hidden p changes e and therefore this marginal balance. It is an initial-state property, not a variable changed by the controller.

Scope. This derivative is exact for the stated analytical member: constant parameters, nonbinding preparation transfer, a common traversal time, homogeneous proportional discharge and fixed active branch. Uneven batch composition, staggered arrivals, changing receiving, or speed-dependent transfer/service parameters require recomputing the population-time integral and derivative. This explanation does not amend the closed contract.

A notation issue in the deck. The ramp slide's $J = \int_0^T P(t)\,\mathrm{d}t$ is *total* system time. The VSL J here is *mean* time. For a fixed total affected mass M_0, the ramp mean and its derivative are obtained by dividing both total quantities by M_0. With fixed mass this does not change optimal actions or gradient signs. The slides now label this distinction.

Source: accepted `s13_m2_model_contract.tex`, Sections 3–6; `s13_m2_certificate_analysis.tex`, gradient section. The exponential law and batch reduction are analytical review assumptions, not a calibrated combined literature model.